

Music Genre Classification using various Machine Learning Algorithms

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Abstract—Music genre classification is a complex task that involves automatically categorizing music tracks into predefined genres. Due to the subjective nature of music and the wide variety of musical elements present within each genre, this task can be challenging. However, machine learning techniques have been increasingly utilized in recent years to enhance the accuracy of music genre recommendation. Music genre classification relies on features such as melody, tempo, pitch, harmony, rhythm, and frequency to categorize music into different genres. In this paper, the GTZAN dataset is being used and to extract features, various methods such as spectrogram, and statistical analysis have been employed. We provide you with an overview of different approaches used in machine learning to achieve the highest level of accuracy using a particular algorithm, indicating its potential for use in music genre classification.

Index Terms—Music Genre Classification, KNN, Decision Tree, Random Forest, Accuracy, Feature Extraction

I. INTRODUCTION

Music genre classification is an essential task in the industry of music recommendation systems or music streaming services. It serves as a means for users to conveniently discover new music based on their preferred genre, and it assists music creators in targeting their intended audience. Despite the various approaches used, such as machine learning and music theory, classifying music tracks into pre-defined genres remains a challenging task due to the subjective nature of music and the diversity of musical elements present within each genre. This study utilizes the GTZAN dataset for music genre classification, which comprises 1000 pre-segmented audio tracks into 30-second clips, each labeled with one of ten genre categories, including blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. Our primary objective is to provide a comprehensive overview of the approaches employed for music genre classification using this dataset. Specifically, we focus on discussing the pre-processing techniques and feature extraction methods such as MFCC, utilized for the dataset to aid in music recommendation and genre classification tasks. Additionally, we provide information regarding the data collection and processing procedures utilized for this study. In the experimental phase of this study, we present a summary and analysis of the result obtained. This includes comparing the performances of various algorithms of machine learning like KNN, Decision Trees, Random Forest, and XGBoost, evaluating the accuracy of the classification and recommendation system, and drawing conclusions from the

findings. Our conclusion provides a comprehensive summary of the key findings and their implications. Furthermore, we provide recommendations for future studies that can build upon the findings of this research to improve the effectiveness and efficiency of music genre classification and recommendation systems.

II. RELATED WORK

In this work, a classification method for 11 Western music genres— including Rock, Pop, Rap, Country, Folk, Metal, Jazz, Blues, R&B, Electronic Music, and Classical Music— was developed. One is to employ two previously trained models. The alternative option is to modify a pre-trained model's last dense layer (MSD_musicnn or MSD_musicnn_big) to output probabilities of 11 genres rather than 50 tags [1]. The idea of ATMGCM with SVM and Random Forest has been adopted in the suggested article since it is more accurate and efficient than the conventional classification approach. While human effort can achieve approximately 70% success rates for genre categorization, our experiment only achieved 61% of that. Genres may be effectively classified using the ATMGCM algorithm. Resnet-34 served as the foundational model for the suggested approach. In addition to feature engineering for a soundwave, the baseline models used were Support Vector Machine (SVM) and Random Forest (RF) [2]. They employed the following techniques for feature extraction using machine learning processing: the derivative of the Mel Frequency Cepstral Coefficients (MFCC), Zero Crossing Rate, spectral bandwidth, roll off, centroid [3] [4], contrast, flatness, RMS (Root Mean Square Energy), polygonal features, chroma CENS, chroma CQT, chroma STFT, tonnes, wavelet, etc. MLP Classifier, Logistic Regression, Random Forest, Linear Discriminant Analysis, K-Neighbors, SVM, Naive Bayes, Gradient Boosting, and Ada Boost Classifiers were used to categorize the data for the classification section [3]. The goal of the acoustic characteristics is to categorize musical genres and give musical suggestions. Digital signal processing in the time and frequency domains is evaluated. For feature extraction, statistical descriptors like mean, median, standard deviation, etc. [4]. The GTZAN dataset from the Music Analysis, Retrieval, and Synthesis for Audio Signals (MARSYAS) website was used for this paper. Python librosa package [6] was used to modify the input audio file into a frequency-time domain signal. With a prediction accuracy of 81% and a classification

accuracy of 67%, the Decision Tree algorithm can be used to classify the genres that the KNN cannot effectively classify [5]. It can assist in acquiring various statistics, such as the genre of music that is popular in a specific area. In their effort, we are accurately and automatically categorizing unidentified music into its genre. Linguistic content is separated from background noise while feature extraction. This aids in achieving a high level of forecast accuracy. In order to focus more on the linguistic substance of the music and less on noise, we transform the original music into a mel-spectrogram. Librosa Python Library [5] was employed [6]. MFCC are frequently used in the classification of musical genres as feature vectors in SVM. Summary of N samples, represented by y , and audio files of M samples, represented by x , is the instance of audio files where their segments matched their representative features. The suggested method simply replicates the steps of the traditional method to pull out summaries from various audio file sections [7]. The efficacy of music genre identification based on data is investigated and tested using three different classifiers: the SVM with radial kernel base function RBF, KNN and Naive Bayes (NB). It was chosen because each song in the Spotify music collection offers thorough information. The SVM classifier, which scored 80% in the testing, performs best in terms of classification accuracy, followed by KNN and NB, which scored 77.18% and 76.08%, respectively [8]. Mel spectrums are obtained from transferring source audio file using librosa. On the basis of Mel spectrum CNN model is trained. Majority voting is used between the ten classifiers to make judgment and the average accuracy recorded on the GTZAN dataset was 84% [9]. MFCC and log energy with dynamic frame duration are the music genre characteristics considered in this study. The dynamic music frame analysis will be used to train an SVM-based music genre classifier with an optimized non-linear decision rule. Analysis will be used to train an SVM-based music genre classifier with an optimized non-linear decision rule. Using six distinct music genres—blue, Bossa, classic, piano, lullaby, dance—the suggested novel music genre classification algorithm was able to obtain 98% of average classification accuracy [10].

III. PROPOSED METHODOLOGY

In this section of music genre classification, the approach used to classify music into different genres is described in detail (refer fig.1). This involves a series of processes like audio data processing, feature extraction, and training of machine learning models for classification. The preprocessing step involves the preparation of audio data before feature extraction. This can include tasks such as audio normalization, noise reduction, and trimming silent portions. Feature extraction is the method of converting the raw audio input into a set of features that can be utilized for machine learning. Common features used for music genre classification include Mel-frequency cepstral coefficients, Spectral features, and Rhythm features. The selection of appropriate features depends on the problem domain and the characteristics of the audio data. The models used for music genre classification may include

traditional classifiers such as Decision Trees, Random Forests, K- Nearest Neighbors (KNN), and XGBoost. These models use the extracted features to learn patterns in the data and make predictions about the genre of a given audio file. The model is tested with appropriate measures like accuracy, precision, recall, and F1 score. To avoid overfitting and robustness of the models, techniques such as Cross- validation are used.

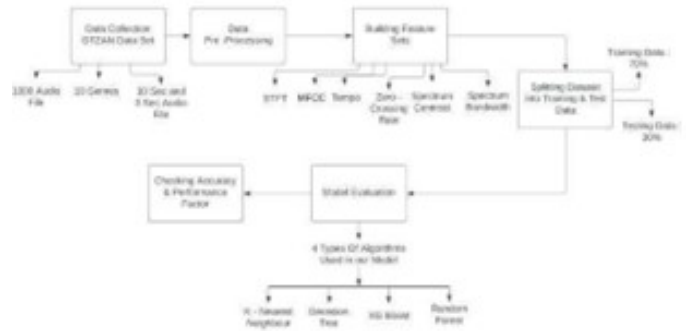


Fig. 1. This figure shows the overview flow of the music genre classification

A. Data Collection

First, we used a dataset from Kaggle which consisted of music tracks, including audio features such as tempo, key, and duration, along with their corresponding genre labels. We have used the most famous dataset for audio classification i.e. GTZAN by G.Tzanetakis. Thousand tracks totaling thirty seconds each, 22050 Hz sample frequency, and 16 bits are included. Rock, blues, reggae, classical, pop, country, metal, disco, hip-hop, jazz are the genres provided in GTZAN. Genres are original - A collection of 10 genres (as shown in fig.2)consisting of hundred audio files each, 30 seconds long CSV files - Including audio file features. With all other features, mean and variance are also calculated and stored in one file,one for individual audio.

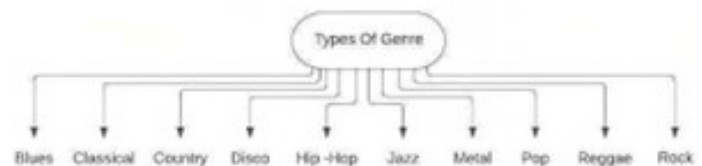


Fig. 2. This figure shows the various genres present in the GTZAN

B. Data Preprocessing

In this paper, data preprocessing techniques were employed to prepare the music dataset for genre classification. The raw audio files were initially converted into a suitable format, such as waveform or spectrogram representation, to extract meaningful features for analysis. The audio signals were then preprocessed to remove any noise or artifacts using techniques like filtering and denoising. Additionally, normalization was applied to ensure consistent scaling across different features.

Feature extraction methods, like Mel- frequency cepstral coefficients (MFCCs) or spectral features, were used to get relevant data from the audio signals. These features were further processed by techniques like dimensionality reduction (e.g., Principal Component Analysis) to decrease computational complexity and eliminate redundant information.

C. Feature Extraction

The task of classifying music tracks into several genres based on their acoustic qualities such as melody, rhythm, timbre and harmony, is known as music genre classification. In this we extract relevant and discriminative features of audio signals which are used to represent the music tracks. To extract these features we are using librosa and python library. Short-term Fourier Transform (STFT) is used for feature extraction, especially for computing the STFT of an audio file and obtaining the matrix of STFT coefficients. STFT reduces the audio file into shorter length which are equally divided and then calculate the Fourier transform of each of these short segments. We have set the default Fast Fourier Transform window size to 2048 and hop length to 512, for visual representation, refer to figure 3. The hop length determines the number of audio frames between STFT columns and is generally recommended to be less than the FFT window size for good results.

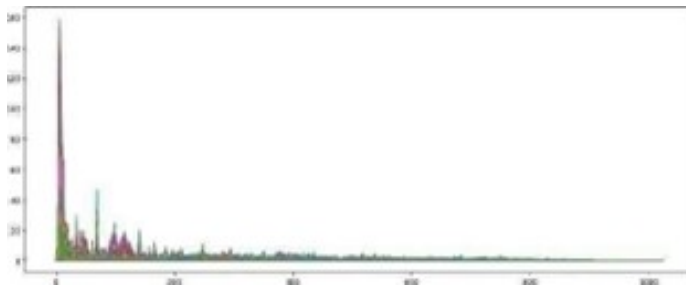


Fig. 3. This figure shows the visual representation of STFT of window size 2048 and hop length 512.

Zero Crossing Rate: The rate at which constant shifting of signal takes place, from positive to zero to negative and vice-versa is called as zero crossing rate. The ZCR can provide information about the overall frequency content of a signal, as well as its rhythmic properties. The sum () function is used to compute the total number of the zero-crossings in the audio signal by summing the elements of the Boolean array which received 39232 zero-crossing rates.

Tempo: Tempo BPM (beats per minute) is how fast or slow a piece of music is performed. The tempo feature tells us the rhythm of a song. The graphical representation can be seen in figure 4. Two parameters are passed to the function and correspond to the audio signal and its sampling rate, respectively. The function returns two values: the estimated tempo in BPM, and an array of beat frame positions in the audio signal. The tempo variable contains the estimated tempo in BPM which we received 107.66 BPM.

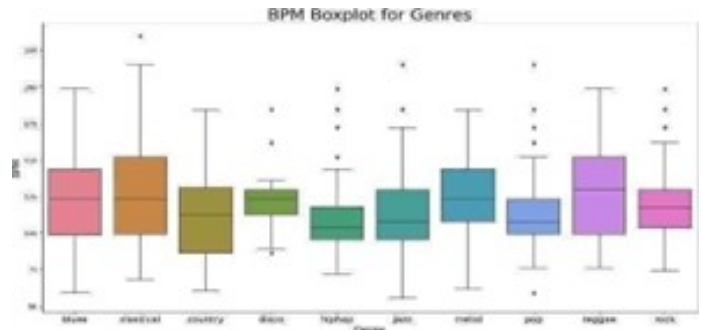


Fig. 4. This figure shows the visual representation of the boxplot.

Mel-Frequency Cepstral Coefficients (MFCC): In studies of speech and voice recognition, these cutting-edge features are employed. The audio signals are first divided into smaller frames because they are continually changing. It takes 20 to 40 milliseconds every frame. Then we attempt to determine various frequencies present in each frame. We will show you the MFCC of various songs which will be trimmed to remove any silent portions of the audio file by identifying and discarding any portions with near-zero amplitude (as shown in figure 5,6 and 7) is then converted to decibels (dB), with the maximum value of the spectrogram as a reference value. In Mel Spectrogram you can see the difference between fig.5,6,7 it is because of the different high and low beats in particular samples. For Metal Mel spectrogram (Fig 5) is dense with purple lines because of continuous high beats, but in classical Mel Spectrogram (Fig 6) the lines are scattered as classical audio is slow and comes on high beats after few intervals. In hip-hop Mel Spectrogram (Fig.07) the lines are continuous but thin as the hip-hop audio comes on high beat continuously.

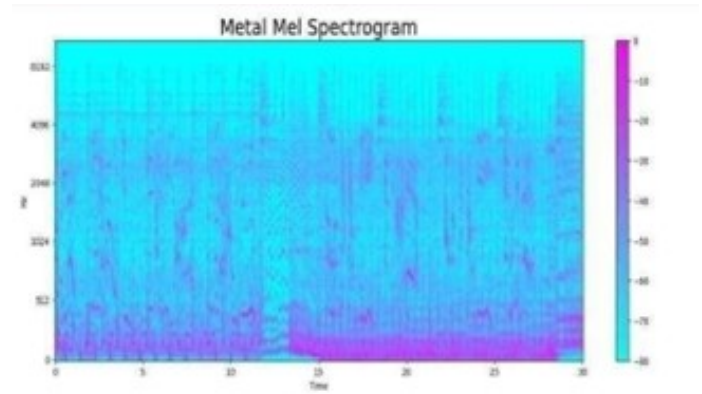


Fig. 5. This figure shows the Mel spectrogram for the metal genre

Spectral centroid: The spectral centroid can be calculated for each segment or frame of a music signal and used as a feature for the classification algorithm. The algorithm can learn to classify the clips based on the features and can be tested on a separate dataset of music clips to evaluate their performance. The spectral centroid can be particularly

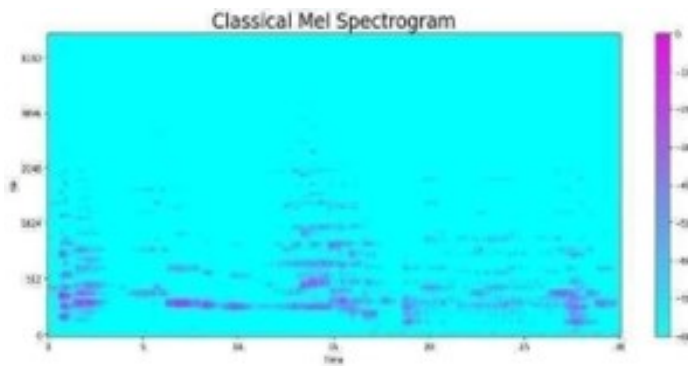


Fig. 6. This figure shows the Mel spectrogram for classical genre

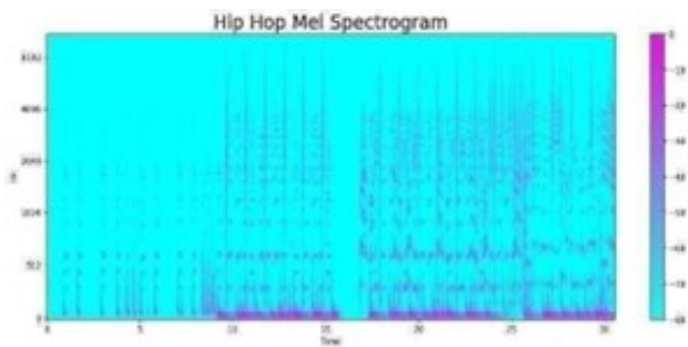


Fig. 7. This figure shows the Mel spectrogram for hip hop genre.

useful in distinguishing between genres that have distinct tonal characteristics, such as classical music, which tends to have a high spectral centroid, and heavy metal, which tends to have a lower spectral centroid.

Spectrum Bandwidth: Spectrum bandwidth refers to the range of frequencies within the electromagnetic spectrum that can be used for communication. In music genre classification, the spectral bandwidth can be calculated for each segment or frame of a music signal and used as a feature for classification algorithms.

D. Data Splitting

To ensure that the machine learning model is performing well on new data and to avoid overfitting, we split up the dataset into training and testing sets. The training set is for training the model and the testing set is used establish the results. 30% of the data will be assigned to the testing set, with the remaining 70% used for training. To maintain consistency in the split between training and testing sets, we are setting a random number generated meaning each time the code is executed with the same random seed value of 42 for better results.

E. Model Selection

Decision Trees method constructs a tree-like model that, using information about a music track's pace, key, and duration, guesses the genre of the song. Decision trees are capable of handling both categorical and continuous data, which are

typically present in music genre categorization difficulties. The system's versatility allows it to forecast genre based on a range of information, such as musical components and acoustic properties. The K-Nearest Neighbors (KNN) method determines a music file's k closest neighbors in a feature space and uses their labels to infer the genre of the track. The 'n- neighbors' option of the KNN classifier object created by this technique is set to 19, which instructs the KNN algorithm to classify test samples based on their 19 closest neighbors. A forecast is made using the Random Forests technique by merging the outputs of numerous decision trees. Random Forest takes use of the variety of the trees by merging several decision trees and averaging their forecasts to increase overall accuracy. The capacity of Random Forest to handle high- dimensional data is another benefit. Extreme Gradient Boost, or XGBoost, is a well-liked machine learning technique used for supervised learning tasks including regression and classification. To build a powerful ensemble model, it combines the advantages of gradient boosting and regularization approaches. The predicted performance of XGBoost can be helpful in music genre categorization, where precisely differentiating between genres is essential .

F. Model Evaluation

We assessed the performance of four chosen models on the testing set, examining metrics like model accuracy. We also utilized visualization tools, such as confusion matrices, to gain insights into model performance and identify potential areas for improvement. Bootstrap Aggregating, also referred to as bagging, is a machine learning ensemble technique aimed at improving the stability and accuracy of statistical classification and regression algorithms. It reduces variance and helps mitigate overfitting, often employed in decision tree methods. Bagging involves training multiple base classifiers on different subsets of the training data, where each classifier is trained independently and makes individual predictions. The final ensemble prediction is typically determined by combining the predictions from all base classifiers, either through majority voting in classification tasks or averaging in regression tasks. TABLE I: This table shows the evaluated results of performance factors of the algorithm(s). The following accuracies are in decimal values of the accuracies of these models evaluated indicating the percentage of the instances in the dataset.

TABLE I
THIS TABLE SHOWS THE EVALUATED RESULTS OF PERFORMANCE FACTORS OF THE ALGORITHM(S). THE FOLLOWING ACCURACIES ARE IN DECIMAL VALUES OF THE ACCURACIES OF THESE MODELS EVALUATED INDICATING THE PERCENTAGE OF THE INSTANCES IN THE DATASET.

Name	Accuracy	After Bagging
KNN	0.80581	0.93333
Decision Tree	0.64364	0.91111
XGBoost	0.89723	0.91112
Random Forest	0.80125	0.92222

G. Data Visualization

We have used Principal Component Analysis as shown in Figure 8, to determine the primary direction of variation within the dataset. Subsequently, we projected the data onto a reduced-dimensional space defined by these identified directions. The dataset separates the features (X) from the target variable (y). Then, the MinMaxScaler class from the preprocessing module is used to normalize the feature values to a range between 0 and 1. Next, The PCA help us to decrease the dimensionality by two dimensions. The resulting principal components are stored in the variable, and a new DataFrame is made to hold them, with the column names "principal component 1" and "principal component 2". Finally, the target variable (y) is concatenated with the principal components DataFrame to create a new DataFrame (finalDf) that contains the reduced feature set along with the target variable. The output shows the percentage of variance explained by each of the two principal components. In this case, the first component has 26.7% of the variance, and the second component has 18.26% of the variance, for a total of 44.93%.

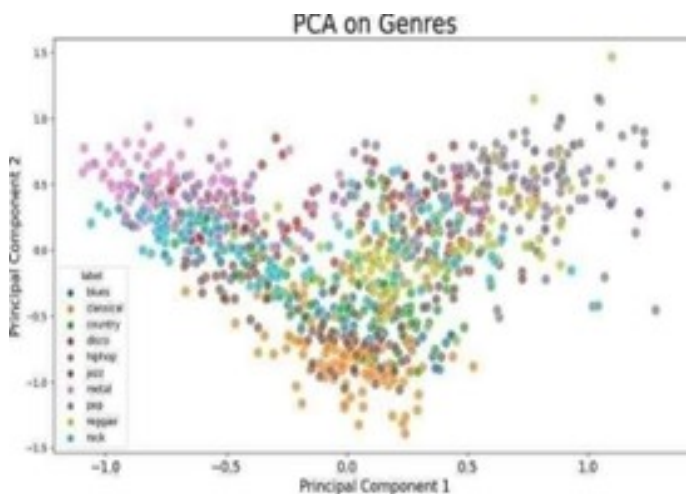


Fig. 8. This figure shows the visualization of Principal Component Analysis

A correlation heatmap visually represents the correlations between various features within a dataset. It displays the pairwise relationships between the features in a graphical format, as shown in Figure 9. The generated heatmap illustrates the relationships between the different features in the dataset. It uses warmer colors to indicate stronger positive correlations and cooler colors to represent stronger negative correlations. By analyzing the heatmap, we can identify which features are most strongly correlated with each other and with the target variable, which can be useful for feature selection and model building.

IV. DISCUSSION

The result of our analysis showed that XGBoost was the most effective algorithm for music genre classification. This algorithm was able to capture the relationships between the audio and the genres, leading to better performance compared

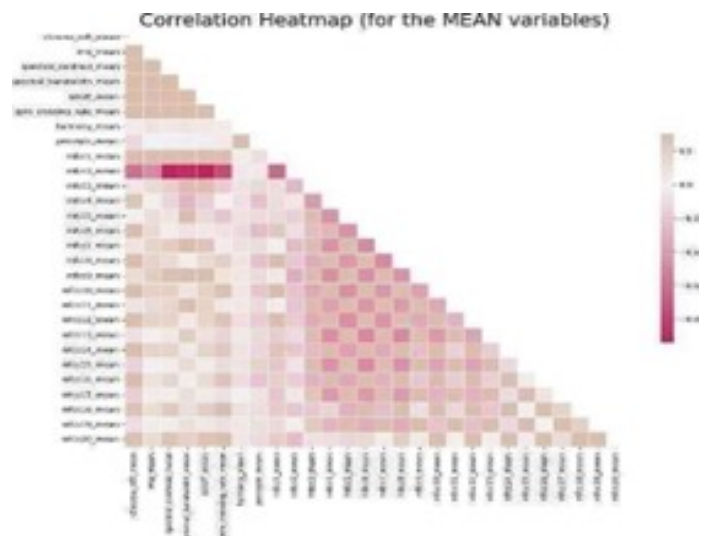


Fig. 9. This figure shows the visualization of the Correlation heatmap.

to the other two algorithms (KNN and decision tree). The decision tree algorithm also performed well, indicating that it may be a good choice for simpler classification tasks. However, the KNN algorithm lagged behind the other two algorithms, suggesting that it may not be the best choice for this particular task. Accuracy can be increased by neural networks or deep learning methods. After successfully performing bagging and evaluating the accuracies of the four models. But after bagging KNN showed the maximum accuracy amongst four models. The estimated neighbors kept for bagged KNN was 10.

V. CONCLUSION

The research paper primarily centers around music genre classification and employs a dataset consisting of 10 distinct music genre folders. Every folder will contain hundred music files according to the respective genre. Using this dataset, the models are trained in 3:1 ratio. Several supervised machine learning algorithms were tested, including XGBoost, KNN, and Decision Tree algorithms, and these were found to perform better than other algorithms in terms of accuracy. Efficiency is a key consideration when evaluating the success of the classification system. The XGBoost algorithm was found to be the most accurate, achieving a test accuracy of 89.72%. In addition to XGBoost, KNN, and Decision Tree algorithms were also used, achieving accuracy scores of 80.58% and 63.46% respectively. A Random Forest algorithm was also tested, achieving an accuracy score of 81.41%.

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